RSCL — Algorithm 4: Target / Address Resolution with Collectives, Exclusions and Ambiguity Refusal

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Date: 2026-11-12 · Document type: formal algorithm specification of the implemented code (HU-MA'AT drone-fleet simulation demo)

Source files described (content hash and state at document build time):

`hu-telepresence-gazebo/festival/mission_intents.py`	`sha256 e109c5a9e89c`	2026-11-12 01:11 UTC	clean @ 05172be
`hu-telepresence-gazebo/festival/hu_live_server.py`	`sha256 2416370e93dc`	2026-11-11 06:38 UTC	clean @ 10ee643

1. Purpose and context

Steering the wrong aircraft is a primary hazard of spoken fleet control. This algorithm computes, from the utterance alone, the exact set of addressed aircraft, or refuses with an explanation. It is used by the deterministic mission router (Algorithm 3) and, in a closely related form (`stated_targets`), as the named-address guard on the neural route (Algorithm 1). Neither form ever infers an aircraft that is not literally named or implied by a collective word.

2. Inputs and outputs

Input: lower-cased, whitespace-normalised utterance *s*; fleet *N* = (Horus, Ra, Thoth, Anubis, Osiris) in fixed order. **Output of** `addresses(s)`: a pair (*targets*, *problem*) where exactly one is non-empty; targets are returned in fleet order. **Output of** `stated_targets(u)`: a list, or None meaning "no explicit address stated".

3. Algorithm

Algorithm 4.1 ADDRESSES (mission route)

Source: festival/mission_intents.py: addresses, COLLECTIVE, EXCLUSION

```
function ADDRESSES(s)
  C ← s contains \b(all|everyone|everybody|the fleet|the swarm|all drones|all five)\b
  X ← first match of \b(except|excluding|but not)\s+NAME(\s*(,|and|&)\s*NAME)*\b   // one or more names
  Named ← [ n ∈ N : whole word lower(n) occurs in s ]
  if s contains \b(or|maybe|perhaps|probably|either)\b then return ([], "Ambiguous target wording; restate one definite aircraft set.")
  if C then
    if Named ≠ ∅ and X = None then return ([], "Do not combine ALL with individual names unless using EXCEPT.")
    if s contains (except|excluding|but not) and X = None then return ([], "Unrecognized exclusion. Say \"all except Ra\".")
    out ← { names occurring inside X }  (∅ if X = None)
   if ∃ n ∈ Named : lower(n) ∉ out then return ([], "Do not combine ALL with individual names unless using EXCEPT.")
   T ← [ n ∈ N : lower(n) ∉ out ]                // fleet order
   if T = ∅ then return ([], "That exclusion leaves no aircraft.")
   return (T, None)
  if X ≠ None then return ([], "Exclusion requires ALL, e.g. \"all except Ra\".")
  if Named ≠ ∅ then return (Named, None)
  return ([], "Specify a named aircraft or ALL.")
```

Algorithm 4.2 STATED-TARGETS (exact-set guard on the neural route)

Source: festival/hu_live_server.py: stated_targets; Service.command

```
function STATED-TARGETS(u)
  Named ← { n ∈ N : whole word lower(n) in lower(u) }
```

```
  C ← lower(u) matches \b(everyone|everybody|all (the )?drones?|all (?=take
off|land|stop|hover|move|rise|ascend|descend))\b
  X ← first \b(except|excluding|but not) NAME\b      // a single name only
  if C then
    if an exclusion word is present and X = None then return None        // abstain; model band stands
    return [ n ∈ N : X = None or lower(n) ≠ X.name ]
  if Named ≠ ∅ then return [ n ∈ N : n ∈ Named ]
  return None
// use: if STATED-TARGETS(u) ≠ None and set(model targets) ≠ set(STATED-TARGETS(u)) then band ← REFUSE
```

4. Flowchart and state diagram

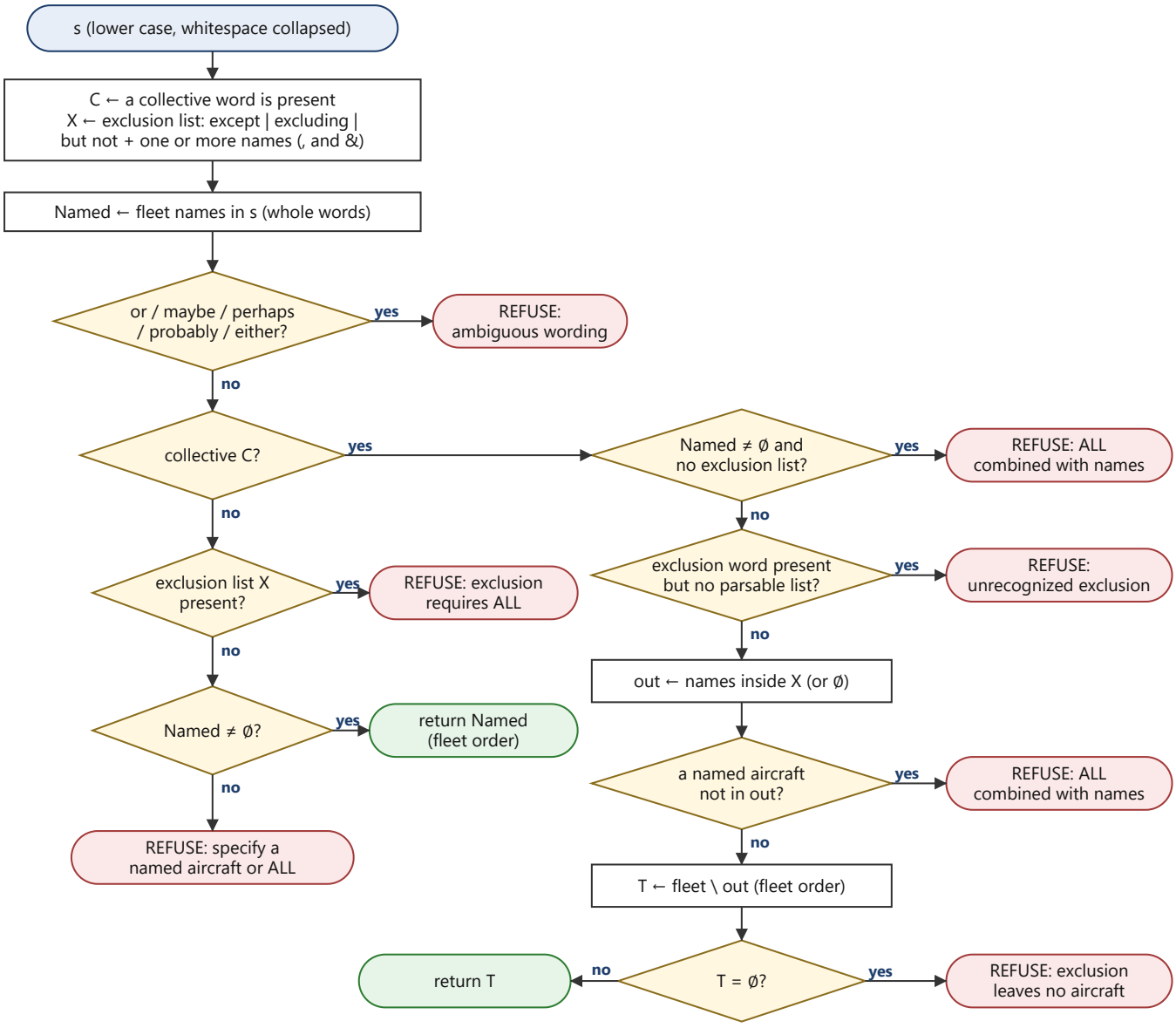


Figure 4.1. Flowchart of `mission_intents.addresses` (Algorithm 4.1). Each REFUSE carries the corrective message quoted in the code. The hedge-word test runs on the whole sentence before anything else.

Address resolution is stateless (no dialogue memory); its data flow is shown instead of a state machine.

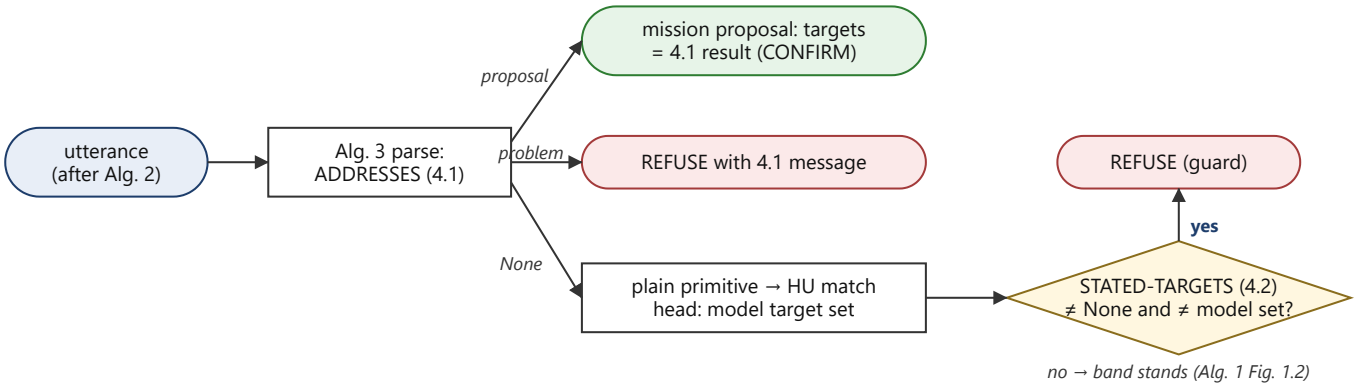


Figure 4.2. Both resolvers are stateless functions of the utterance (no dialogue memory, no telemetry). 4.1 decides the targets of mission proposals; for plain take-off / land / hover / stop sentences, which go to HU, only 4.2 is applied, as a consistency guard on the model's own target set.

Table 4.1. Outputs computed by calling `addresses`, `parse` and `stated_targets` directly on the current source (no aircraft involved). Where the route is HU, the dispatched set is the model's; 4.2 only decides whether that set is refused.

Utterance	ADDRESSES (4.1)	Route (Alg. 3)	STATED-TARGETS (4.2)
Everyone except Ra, take off	Horus, Thoth, Anubis, Osiris	HU (plain primitive)	Horus, Thoth, Anubis, Osiris — agrees
All except Ra and Thoth, take off	Horus, Anubis, Osiris	HU (plain primitive)	Ra, Thoth — **differs from 4.1**
All drones except Ra, Thoth, land	Horus, Anubis, Osiris	HU (plain primitive)	Horus, Thoth, Anubis, Osiris — **differs from 4.1**
Everyone except Ra and Thoth, land	Horus, Anubis, Osiris	HU (plain primitive)	Horus, Thoth, Anubis, Osiris — **differs from 4.1**
All except Ra and Thoth, move up 50 cm	Horus, Anubis, Osiris	mission proposal (CONFIRM): move	not applied
All except Ra, Thoth and Osiris, form a V	Horus, Anubis	mission proposal (CONFIRM): formation	not applied
Everyone except Horus, Ra, Thoth, Anubis and Osiris, land	refused: That exclusion leaves no aircraft.	REFUSE	not applied
All except Ra form a V, Thoth	refused: Do not combine ALL with individual names unless using EXCEPT.	REFUSE	not applied
Horus or Ra, take off	refused: Ambiguous target wording; restate one definite aircraft set.	REFUSE	not applied
Ra and Thoth, take off	Ra, Thoth	HU (plain primitive)	Ra, Thoth — agrees
All except, take off	refused: Unrecognized exclusion. Say "all except Ra".	REFUSE	not applied
Land	refused: Specify a named aircraft or ALL.	REFUSE	not applied

5. Parameters

Vocabulary	Mission route (4.1)	Neural guard (4.2)
Collective	all, everyone, everybody, the fleet, the swarm, all drones, all five	everyone, everybody, all (the) drone(s), "all" directly before take off / land / stop / hover / move / rise / ascend / descend

Exclusion	except / excluding / but not + one or more fleet names separated by commas, *and* or *&*; every other named aircraft must be inside the list	except / excluding / but not + the first name only
Hedge words (refuse)	or, maybe, perhaps, probably, either	none (guard abstains)
Name matching	exact whole-word, case-insensitive	same; voice aliases applied earlier (Alg. 19)

6. Complexity

O(|N|·|s|) for the name scans plus a constant number of regular-expression searches.

7. Properties

Guaranteed

- **No inferred aircraft:** a target is returned only if named, or implied by a collective minus a named exclusion.
- **Ambiguity refusal:** any hedge word refuses the address (even if it appears elsewhere in the sentence; this is conservative).
- Mixed forms ("all, Horus") and orphan exclusions ("except Ra" without ALL, or "all except" without a recognised name) are refused with a corrective message.
- On the neural route, a model target set that differs in any element from an explicitly stated address is refused, never repaired.
- An exclusion that removes every aircraft is refused ("That exclusion leaves no aircraft.").
- Deterministic and order-stable output (fleet order).

Not guaranteed / limitations

- **The two resolvers disagree on exclusion lists.** 4.1 removes every listed name ("all except Ra and Thoth" → Horus, Anubis, Osiris). 4.2 removes only the first listed name, and it treats plain *all* as collective only immediately before a listed verb. Plain take-off / land / hover / stop sentences are routed to HU (Algorithm 3), so for them only 4.2 is applied: for "All except Ra and Thoth, take off" it returns {Ra, Thoth}, the aircraft the operator excluded. A correct model prediction is then refused (safe direction); a model prediction of {Ra, Thoth} would pass the guard and could be dispatched. For "Everyone except Ra and Thoth, land" 4.2 returns the fleet minus Ra, so a correct prediction is refused. See Table 4.1. **Not covered by a test.**
- The two vocabularies differ: e.g. "the fleet" is collective in 4.1 but not in 4.2; plain "all" is collective in 4.1 but in 4.2 only before listed verbs. Where 4.2 returns None it abstains.
- Ordinal/indexed addressing ("drone two") is not resolved here (the HU model has an index feature, Algorithm 1).
- Name matching is literal; mis-recognised names are handled only by the voice alias table and its confirmation step (Algorithm 19).

8. Testing and measured results

- `acceptance/tests/test_v9.py::test_reject_unknown_ambiguous_conflicting_and_unsafe`, `test_wrong_model_label_refused_even_high_confidence`; `test_missions.py::test_turn_target`.
- Live command matrix (`run_20261111T052914Z`): "Horus or Ra, take off" → refused (ambiguous); "All except, take off" → refused (unrecognised exclusion); "land" → refused (no address); "Everyone except Ra, take off" → Horus, Thoth, Anubis, Osiris. Fault scenario: "All except Anubis form a V shape" → four aircraft (PASS).
- The disagreement between 4.1 and 4.2 for exclusion lists is **not covered by a test**; Table 4.1 was computed by calling the functions directly. The live command matrix above predates multi-name exclusions.

9. Comparison with alternatives and related work

Deciding which aircraft an utterance refers to is a reference-resolution problem. Established work resolves references from discourse context, from learned models, from perception, or by asking.

- **Discourse-based resolution (centering)** [1]: an attentional state carried across utterances explains which entity a pronoun or reduced expression refers to.
- **Neural coreference resolution** [2]: every span is scored as a possible mention and a distribution over antecedents is learned end to end.
- **Interactive grounding with clarifying questions** [3]: candidates are scored against the referring expression and, when the reference is ambiguous, the robot asks a generated disambiguating question.
- **Selecting and grouping robots by attention and speech** [4, 5]: the user looks at robots to select them; in [5] the user says how many ("you two"), forms a team that is then commanded as a unit ("take off"), and modifies it ("not you").

Approach	Evidence used	Context / memory	When unsure	This resolver, by comparison
Centering [1]	discourse structure	yes, across utterances	ranked preference	no memory: "them", "it", "the same two" are not resolved (refused)
Neural coreference [2]	learned span features	document	most probable antecedent	no learning; exact names only
INGRESS [3]	vision + language	current scene	asks a targeted question	refuses with a corrective message instead of asking
Face engagement / "You two!" [4, 5]	gaze, speech, counts	persistent teams	election among robots	no deixis, counts or persistent groups; no perception uncertainty
This document	literal names, collective words, exclusion lists	none	refuses (hedge words, mixed forms, empty set)	—

Where this resolver is weaker. It handles only the closed fleet of five literal names, has no notion of previous commands, of deictic or numeric selection, or of groups, and its two variants disagree on exclusion lists (Section 7). It never asks a specific question; the operator must restate the command. **Where it differs.** Its output is exact and explainable: no aircraft is ever inferred, and any hedge word causes refusal, which is the conservative choice for actuating vehicles; the methods above optimise resolution accuracy rather than refusal of uncertain references.

10. Relation to other algorithms

Called by Algorithm 3 for every mission branch; the neural form is a guard step of Algorithm 1. The resulting non-participants become fixed obstacles in Algorithm 5 (`assign_shifted`) and stationary peers in Algorithm 8.

11. References

1. B. J. Grosz, A. K. Joshi and S. Weinstein, "Centering: A Framework for Modeling the Local Coherence of Discourse," *Computational Linguistics*, 21(2):203–225, 1995. https://aclanthology.org/J95-2003
2. K. Lee, L. He, M. Lewis and L. Zettlemoyer, "End-to-end Neural Coreference Resolution," *EMNLP* 2017, pp. 188–197. https://aclanthology.org/D17-1018
3. M. Shridhar and D. Hsu, "Interactive Visual Grounding of Referring Expressions for Human-Robot Interaction," *Robotics: Science and Systems XIV*, 2018. https://doi.org/10.15607/RSS.2018.XIV.028
4. A. Couture-Beil, R. T. Vaughan and G. Mori, "Selecting and Commanding Individual Robots in a Vision-Based Multi-Robot System," *Canadian Conference on Computer and Robot Vision (CRV)*, 2010. https://www2.cs.sfu.ca/~mori/research/papers/selectingandcommanding_crv.pdf
5. S. Pourmehr, V. Monajjemi, R. Vaughan and G. Mori, "'You two! Take off!': Creating, modifying and commanding groups of robots using face engagement and indirect speech in voice commands," *IEEE/RSJ IROS* 2013, pp. 137–142.

https://www2.cs.sfu.ca/~mori/research/papers/pourmehr-iros13.pdf

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This specification describes the behaviour of the source revision identified above. Line-level references use `file:function`. Where the code and its own comments disagree, the code is described. Simulation-only system: no statement here is a claim about flight hardware.